

Absolute Value Inequalities: Special Cases and Interval Notation

Algebra II | Unit 2 | Day 8 | TEK A2.6.F

Objective: I can write absolute value inequality solutions in interval notation, solve harder cases, and recognize when an inequality has no solution.

Vocabulary

No solution — NO value of x works. Happens when an isolated absolute value is set ____ or ____ a negative number.

Part 1 — Interval Notation Reference

Inequality	Compound form	Interval notation
$ x > 5$	_____	_____
$ x \geq 5$	_____	_____
$ x < 5$	_____	_____
$ x \leq 5$	_____	_____

Part 2 — Variable on Both Sides

Example. $|3x + 1| - 4 \leq 2x - 1$

Isolate: $|3x + 1| \leq$ _____

$-(2x + 3) \leq 3x + 1 \leq 2x + 3$

Solve both halves and combine: _____

Part 3 — When There Is No Solution

Example. $4 + |x + 1| \leq 2$

Isolate: $|x + 1| \leq$ _____

An absolute value can never be negative, so it can never be $\leq$ a negative number. Solution:

Practice

- Write $|x| \geq 8$ in interval notation. _____
- Write $|x| < 3$ in interval notation. _____
- Solve $|x - 2| + 6 \leq 1$ _____
- Solve $|2x - 1| > x + 5$ _____

Exit Ticket

Solve $3 + |x - 4| < 1$, and explain why the answer comes out the way it does.
